UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 3

Fall 2025

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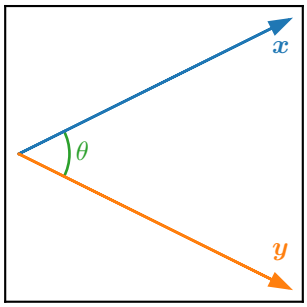
An example of inner product on $\mathbb{R}^n$ is

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \beta_1 x_1 y_1 + \beta_2 x_2 y_2 + \cdots + \beta_n x_n y_n,$$

where $\beta_1, \beta_2, \ldots, \beta_n$ are positive constants. If we pick $\beta_1 = \beta_2 = \cdots = \beta_n = 1$, we obtain the standard inner product that we use in this class:

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^{\mathrm{T}} \boldsymbol{y} = \boldsymbol{y}^{\mathrm{T}} \boldsymbol{x} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

Note that: $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \|\boldsymbol{x}\|_2 \cdot \|\boldsymbol{y}\|_2 \cdot \cos(\theta)$, where θ is the angle between vectors $\boldsymbol{x}$ and $\boldsymbol{y}$.



If $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$, then $\theta = 90^\circ$ and the two vectors are orthogonal.

Example 1. Given two articles published by two news outlets CNN and Fox News, we want to study their similarity.

- Define a simplified dictionary of major keywords on the topics of the articles, such as:

$$\{\text{president}, \text{tax}, \text{middle east}, \ldots\}.$$

- Define $\boldsymbol{x}$ as a vector in $\mathbb{R}^n$, where x_i is the number of times word i in dictionary appears in article 1.
- Define $\boldsymbol{y}$ similarly for article 2.
- Calculate $\cos\theta$.
- A value close to 0 shows articles are uncorrelated while close to 1 shows correlation.

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- Two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ are *orthogonal* if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$, denoted as $\boldsymbol{x} \perp \boldsymbol{y}$.
- d vectors $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)}$ are *mutually orthogonal* if $\langle \boldsymbol{x}^{(i)}, \boldsymbol{x}^{(j)} \rangle = 0, \ \forall i, j \in \{1, \ldots, d\}$ and $i \neq j$.
- If $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)}$ are mutually orthogonal, then they are linearly independent.

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- d vectors $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)}$ are *orthonormal* if

$$\langle \boldsymbol{x}^{(i)}, \boldsymbol{x}^{(j)} \rangle = \begin{cases} 0, & \text{if } i \neq j, \\ 1, & \text{if } i = j. \end{cases}$$

  In other words, since $\|\boldsymbol{x}\|_2^2 = \langle \boldsymbol{x}, \boldsymbol{x} \rangle$, it means the vectors are mutually orthogonal and have length 1.

From now on, whenever we use $\|\boldsymbol{x}\|$, we mean $\|\boldsymbol{x}\|_2$, unless it is said otherwise.

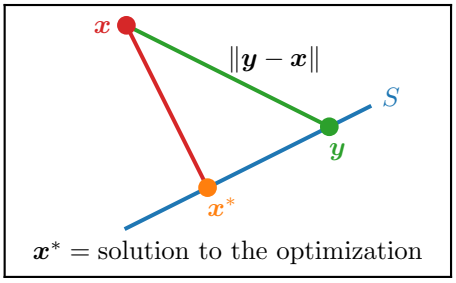
Proposition 1. *If* $\boldsymbol{x} \perp \boldsymbol{y}$, *then* $\|\boldsymbol{x} + \boldsymbol{y}\|^2 = \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2$.

Projections onto subspaces.

- Let S be a subspace of some space X (e.g., $\mathbb{R}^n$ or $\mathbb{R}^{n \times n}$).
- Given a point $\boldsymbol{x} \in X$, the projection of $\boldsymbol{x}$ onto S is shown as $\Pi_S(\boldsymbol{x})$ and defined as:

$$\arg\min_{\boldsymbol{y} \in S} \|\boldsymbol{y} - \boldsymbol{x}\|.$$

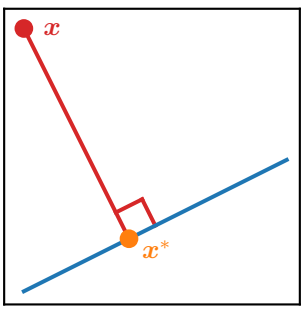
- So, we need to solve optimization $\min_{\boldsymbol{y} \in S} \|\boldsymbol{y} - \boldsymbol{x}\|$ and $\arg\min$ denotes its (unique) minimizer.
- Among all points $\boldsymbol{y}$ in S, we look for the closest one to $\boldsymbol{x}$.



Theorem 1 (Projection onto subspace)**.** *The projection* $\boldsymbol{x}^* = \Pi_S(\boldsymbol{x})$ *exists and is unique.*

A necessary and sufficient condition for $\boldsymbol{x}^*$ to be the projected point is that $(\boldsymbol{x} - \boldsymbol{x}^*) \perp S$, meaning that $\boldsymbol{x} - \boldsymbol{x}^*$ is orthogonal to all vectors in S.

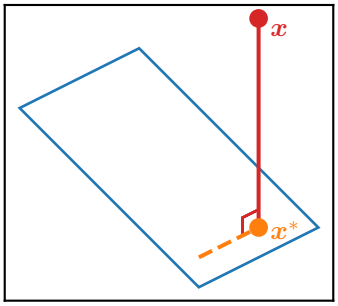
Example 2. In $\mathbb{R}^2$:



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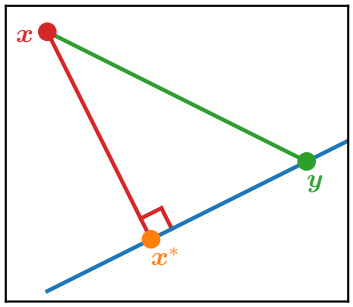
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Example 3. In $\mathbb{R}^3$:



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Note that if $\boldsymbol{x} \in S$, then $\boldsymbol{x}^* = \boldsymbol{x}$.
Intuition behind the proof:

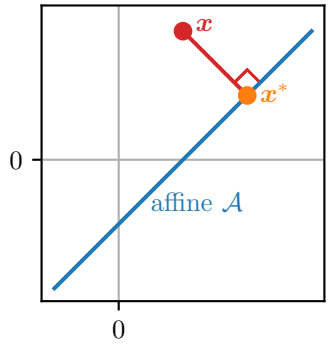


$$\begin{aligned} \|\boldsymbol{x} - \boldsymbol{y}\|^2 &= \|\boldsymbol{x} - \boldsymbol{x}^* + \boldsymbol{x}^* - \boldsymbol{y}\|^2 \\ &= \|\boldsymbol{x} - \boldsymbol{x}^*\|^2 + \|\boldsymbol{x}^* - \boldsymbol{y}\|^2 \qquad (1) \\ &\geq \|\boldsymbol{x} - \boldsymbol{x}^*\|^2, \end{aligned}$$

where (1) is by Proposition 1 and the fact that $(\boldsymbol{x} - \boldsymbol{x}^*) \perp (\boldsymbol{x}^* - \boldsymbol{y})$. Thus we get

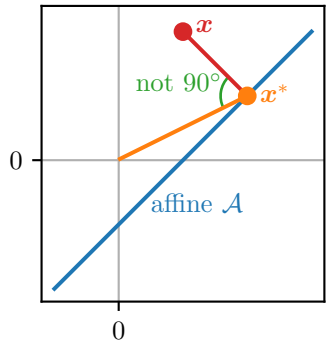
$$\min_{\boldsymbol{y} \in S} \|\boldsymbol{x} - \boldsymbol{y}\| \geq \|\boldsymbol{x} - \boldsymbol{x}^*\|.$$

Also, equality holds if $\boldsymbol{y}$ is chosen as $\boldsymbol{x}^*$.
How about projection on affine spaces?



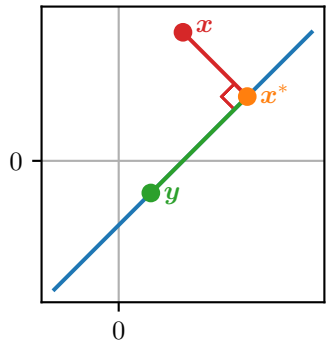
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Note: we cannot say $(\boldsymbol{x} - \boldsymbol{x}^*) \perp \mathcal{A}$ because affine $\mathcal{A}$ is no longer a subspace. For example: $(\boldsymbol{x} - \boldsymbol{x}^*) \perp \boldsymbol{x}^*$ is not true:



(Note: vector $\boldsymbol{x}^*$ goes through the origin and is not orthogonal to $\boldsymbol{x} - \boldsymbol{x}^*$.)
We need to say:

$$(\boldsymbol{x} - \boldsymbol{x}^*) \perp (\boldsymbol{y} - \boldsymbol{x}^*), \ \forall \boldsymbol{y} \in \mathcal{A}.$$



In contrast, for a subspace S, we have $\boldsymbol{y} - \boldsymbol{x}^* \in S$ for any $\boldsymbol{y} \in S$.[1] (Remember: in a subspace, the linear combination of elements belongs to the same subspace: $\boldsymbol{y} \in S, \boldsymbol{x}^* \in S \Rightarrow \boldsymbol{y} - \boldsymbol{x}^* \in S$.) Since $\boldsymbol{y} - \boldsymbol{x}^*$ can be replaced with $\boldsymbol{z} \in S$, we have

$$(\boldsymbol{x} - \boldsymbol{x}^*) \perp \boldsymbol{z}, \ \forall \boldsymbol{z} \in S \quad \Longrightarrow \quad (\boldsymbol{x} - \boldsymbol{x}^*) \perp S.$$

Example 4. Projection onto a one-dimensional subspace.

- Let S be a subspace of dimension 1.
- Let the vector $\boldsymbol{v}$ be a basis for S:

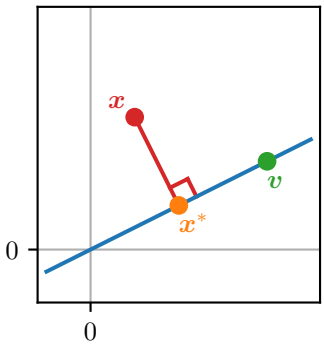
$$S = \{\alpha \boldsymbol{v} \mid \alpha \in \mathbb{R}\}.$$

- Let $\boldsymbol{x}$ be an arbitrary vector in $\mathbb{R}^n$. Let

$$\boldsymbol{x}^* = \Pi_S(\boldsymbol{x}) = \arg\min_{\boldsymbol{y} \in S} \|\boldsymbol{y} - \boldsymbol{x}\|.$$

[1] This is not true for general affine spaces.

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We have $(\boldsymbol{x} - \boldsymbol{x}^*) \perp S$, or equivalently,

$$\langle \boldsymbol{x} - \boldsymbol{x}^*, \boldsymbol{y} \rangle = 0, \ \forall \boldsymbol{y} \in S. \tag{2}$$

Also, since $\boldsymbol{x}^* \in S$, there exists α^* such that $\boldsymbol{x}^* = \alpha^* \boldsymbol{v}$ (since S is one-dimensional).
Set $\boldsymbol{y} = \boldsymbol{v}$ in Equation (2), we get

$$\begin{aligned} &\langle \boldsymbol{x} - \alpha^* \boldsymbol{v}, \boldsymbol{v} \rangle = 0 \\ \Rightarrow\ &\langle \boldsymbol{x}, \boldsymbol{v} \rangle - \alpha^* \|\boldsymbol{v}\|^2 = 0 \\ \Rightarrow\ &\Pi_S(\boldsymbol{x}) = \alpha^* \boldsymbol{v} = \frac{\langle \boldsymbol{x}, \boldsymbol{v} \rangle}{\|\boldsymbol{v}\|^2} \cdot \boldsymbol{v}. \end{aligned}$$

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Example 5 (Example 2.15 from textbook)**.** Consider the dataset of the votes of $n = 100$ U.S. senators on $m = 645$ bills in the period of 2004–2006.
For $j = 1, \ldots, n$, define $\boldsymbol{x}^{(j)} \in \mathbb{R}^m$ as the vector of all votes of senator j:

$$x_i^{(j)} = \begin{cases} 1, & \text{if vote is in favor of bill } i, \\ 0, & \text{if vote is against bill } i. \end{cases}$$

- $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ are high-dimensional data and hard to analyze.
- We project onto one-dimensional subspace to extract information.
- Let $\boldsymbol{v} = [1, 1, \ldots, 1]^{\mathrm{T}} \in \mathbb{R}^m$ denote the direction corresponding to the "average bill."[2]
- Define $\hat{\boldsymbol{x}}$ as the average of all vote vectors:

$$\hat{\boldsymbol{x}} := \frac{1}{n} \left(\boldsymbol{x}^{(1)} + \boldsymbol{x}^{(2)} + \cdots + \boldsymbol{x}^{(n)} \right).$$

- We consider $\boldsymbol{x}^{(1)} - \hat{\boldsymbol{x}}, \boldsymbol{x}^{(2)} - \hat{\boldsymbol{x}}, \ldots, \boldsymbol{x}^{(n)} - \hat{\boldsymbol{x}}$.

[2]This direction encodes the average level of support across all bills: projecting any vector onto $\boldsymbol{v}$ yields the sum of its entries, so a senator who tends to support more bills than average will have a larger projection, and vice versa.

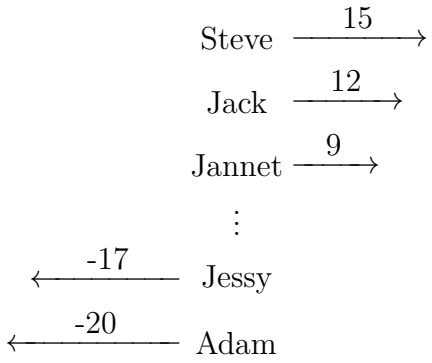
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- The average of these vectors is zero, so $\hat{\boldsymbol{x}}$ removes the bias.
- We project $\boldsymbol{x}^{(j)} - \hat{\boldsymbol{x}}$ (for $j = 1, \ldots, n$) onto the subspace of $S = \text{span}(\boldsymbol{v})$.
- This shows how senator j would vote on the hypothetical average bill $\boldsymbol{v}$ based on their votes on bills $1, \ldots, m$:

$$\Pi_S\left(\boldsymbol{x}^{(j)} - \hat{\boldsymbol{x}}\right) = \boxed{\frac{\langle \boldsymbol{x}^{(j)} - \hat{\boldsymbol{x}}, \boldsymbol{v} \rangle}{\|\boldsymbol{v}\|^2}} \cdot \boldsymbol{v}.$$

This is a scalar. Call it: score on average bill.

- Let's arrange the senators based on their scores from highest to lowest:



- You can group the senators into two groups, one with positive scores and another one with negative scores.
- It happens that the groups almost correctly identify the Republican and Democratic groups (for scores close to zero that might lead to mistakes).
- Here's a Kaggle notebook that you can play around with, which analyzes the senate voting data from 2015 to 2016: https://www.kaggle.com/code/georgemlp2/us-senate-voting-data-2015-2016.

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The previous formula was for projection onto a one-dimensional subspace, and we need to generalize it.
Let S be a d-dimensional subspace. Define a basis $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(d)}$ for S, thus $S = \text{span}(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(d)})$.

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